Present-epoch scale matching, the weak-field invariant $C_{\rm obs}$, and no-go results for a cubic T^3 geometric story

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The coincidence $2\pi a_0 \sim cH_0$ is a standard observation in the MOND literature. Several recent proposals attempt to elevate it to a *geometric derivation* by combining (i) a compact flat cubic three-torus of present-day side $L = c/H_0$ with a putative "circulation quantum" $\kappa = cL$, and (ii) the isotropic projector ratio $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$ with the weak-field coefficient in an AQUAL-type action. We show that neither step is presently a derivation. Writing $\kappa = cL$ does not quantize circulation; the package $L = c/H_0$, $\Gamma = cL$, and $\omega = H_0$ is kinematically inconsistent if L is a circumference; and cubic E_1 topology bounds require $L \gtrsim \mathcal{O}(6)\, c/H_0$, excluding the $\alpha = 1$ matching that reproduces the MOND scale. Exact tracking $L_{\rm phys}(t) = c/H(t)$ is incompatible with fixed comoving moduli in ordinary FLRW except for a coasting expansion. The projector identity is correct but topology-independent and does not appear in the weak-field action, so it cannot fix $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$, which we derive as the unique invariant of field rescaling. Imposing both $a_0 = cH_0/2\pi$ and $C_{\rm obs} = 2/3$ yields $a_{0,\rm eff} = (4/9)a_0$, in conflict with the empirical RAR normalization. What remains is a present-epoch phenomenological encoding of the coincidence, a clean EFT invariant, and an explicit reconstruction checklist for any future geometric claim.

I. INTRODUCTION

Empirical galactic dynamics exhibit a characteristic acceleration $a_0 \sim 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$ that organizes the radial acceleration relation (RAR) and the baryonic Tully–Fisher relation [1–4]. The numerical coincidence

$$2\pi\, a_0 \sim c\, H_0 \tag{1}$$

is well known [1, 2]. It is a legitimate research question whether a compact flat three-torus $T^3 = S^1 \times S^1 \times S^1$ [5, 6] and a weak-field effective field theory (EFT) can turn that coincidence into a derivation.

A recurring two-step proposal answers "yes":

(A) "Circulation quantization" on a domain of size $L = c/H_0$ yields $a_0 = cH_0/2\pi$;

(B) the transverse-projector ratio $2/3$ yields the EFT coefficient $C_{\rm obs}$ in a deep-MOND square-root law.

Both steps fail under standard technical scrutiny. This paper is a short technical note: it withdraws those derivations, records the associated no-go results, and isolates what *does* follow cleanly from a static AQUAL-type action [7]. Table I summarizes the status of each claim.

TABLE I. Status of claims discussed in this work.

Quantity / claim	Status	Note
Empirical coincidence $2\pi a_0 \sim cH_0$	Fact	Motivating observation only
$a_0 \equiv cH_0/2\pi$ at the present epoch	Phen.	Named postulate; not derived
"Circulation quantization $\Rightarrow a_0$"	Withdrawn	$\kappa = cL$ is not h/m quantization
Package $(L, \Gamma = cL, \omega = H_0)$ as prior proposals	Withdrawn	Inconsistent if L is a circumference
Cubic E_1 with $L = c/H_0$ vs. Planck topology	Excluded	$L \gtrsim \mathcal{O}(6)\, c/H_0$ required
$L_{\rm phys}(t) = c/H(t)$ with fixed comoving moduli	Excluded	Needs evolving moduli or coasting FLRW
Isotropic count $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$	Derived	Generic 3D identity; not special to T^3
$C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$	Derived	Field-normalization invariant
$C_{\rm obs} = 2/3$ from the trace ratio	Withdrawn	Projector absent from the action
Simultaneous $a_0 = cH_0/2\pi$ and $C_{\rm obs} = 2/3$ as RAR fit	Excluded	Predicts $a_{0,\rm eff} = (4/9)a_0$
Compensated sources on compact T^3	Derived	$\int \mathrm{div} = 0$
Geometric derivation of a_0 or $C_{\rm obs}$	Open	Path in Sec. VIII

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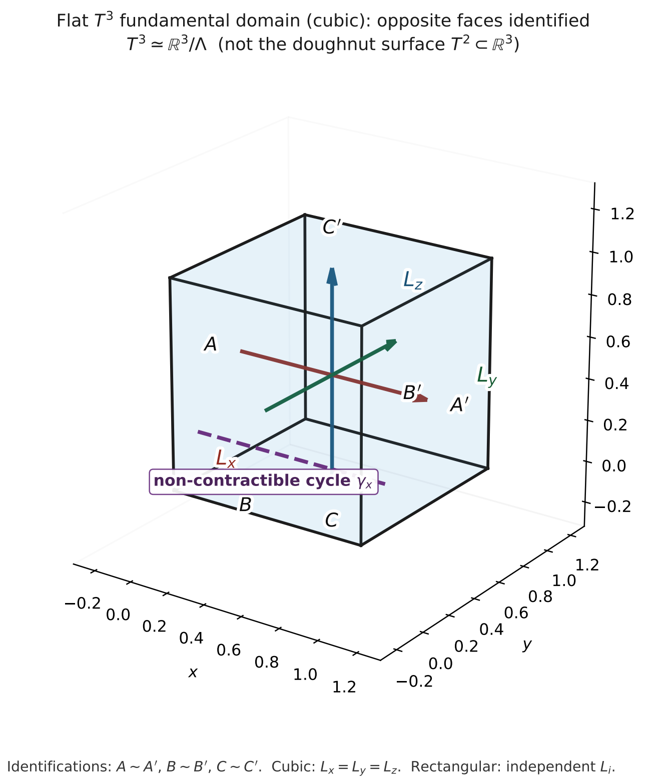

FIG. 1. Flat cubic T^3 fundamental domain: opposite faces identified ($A \sim A'$, $B \sim B'$, $C \sim C'$). Edge lengths L_x, L_y, L_z are the three independent periods. Cubic means $L_x = L_y = L_z$. This is not the embedded doughnut T^2.

II. CUBIC THREE-TORUS GEOMETRY AND NOTATION

A flat three-torus is the quotient $\mathbb{R}^3/\Lambda$ by a lattice of translations [5]. The correct schematic is a parallelepiped with opposite faces identified (Fig. 1), not the doughnut surface of revolution in $\mathbb{R}^3$. That doughnut is an embedding of T^2.

"T^3" alone does not imply equal periods. Isotropic statements require a *cubic* (or explicitly isotropised) lattice. Rectangular and twisted flat manifolds ($E_2, E_3, \ldots$) are different geometries with different CMB constraints [6, 8].

Throughout, L means a *physical* edge length at the epoch under discussion unless "comoving" is written explicitly. When a numerical Hubble scale is needed we take $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for illustration; none of the no-go arguments depends on that choice at the factor-of-two level.

III. $a_0 = cH_0/2\pi$ IS NOT CIRCULATION QUANTIZATION

A. What quantization requires

In a superfluid with order parameter $\Psi = \sqrt{n}\, e^{i\theta}$, the velocity follows from the phase gradient and

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n\,\frac{h}{m}\,, \qquad n \in \mathbb{Z}\,, \tag{2}$$

with single-valuedness of Ψ on closed cycles [9–11]. Equation (2) quantizes *phase winding*, not "length times a speed limit."

B. What the geometric proposals actually write

Those proposals introduce

$$\kappa_{\rm cosmo} \overset{?}{=} c\,L \tag{3}$$

by multiplying a maximum speed by a length. For constant speed on a cycle of length L, one has only $\Gamma = vL$; setting $v = c$ yields $\Gamma = cL$. No integer winding, no order parameter, and no mass m enter. That is a kinematic assignment, not quantization.

If one *did* set the $n = 1$ quantum equal to c^2/H_0,

$$m = \frac{hH_0}{c^2} \sim 10^{-32}\,\mathrm{eV}/c^2\,, \tag{4}$$

the microscopic mass reappears as a sharp prediction—the opposite of claiming that no microphysics is needed.

C. Internal inconsistency of the (L, Γ, ω) package

Suppose $L = c/H_0$ is the physical circumference of a closed cycle, so $R = L/(2\pi)$. Then:

- $\Gamma = cL$ implies $v = c$;
- $\omega = H_0$ with $v = \omega R$ implies $v = H_0 L/(2\pi) = c/(2\pi)$, contradicting $v = c$;
- the centripetal acceleration c^2/R is $2\pi c H_0$, not $cH_0/2\pi$.

Thus one cannot simultaneously maintain

$$L = \frac{c}{H_0}\,, \qquad \Gamma = cL\,, \qquad \omega = H_0 \tag{5}$$

under that geometric reading of L. "Distributing cH_0 across a 2π angular topology" is not coordinate-invariant: the range of an angular chart can be rescaled. A physical 2π arises only from a defined phase convention coupled to velocity.

a. Conclusion.

$$a_0 \equiv \frac{cH_0}{2\pi} \tag{6}$$

may be retained only as a *present-epoch phenomenological scale-matching postulate* that encodes Eq. (1). For $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ one finds $a_0 \approx 1.1 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, of the observed order of magnitude [3]. All "geometric derivation from circulation quantization" language should be withdrawn.

IV. CUBIC TOPOLOGY BOUNDS EXCLUDE $L = c/H_0$

Planck topology analyses constrain cubic E_1 three-tori tightly. In terms of the radius R_i of the largest sphere inscribed in the fundamental domain, Planck reports $R_i > 0.97\,\chi_{\rm rec}$ for the cubic case at high confidence [6]. For a cube of side L, $R_i = L/2$, so

$$L \gtrsim 1.94\,\chi_{\rm rec}\,. \tag{7}$$

With $\chi_{\rm rec} \simeq 3.15\,c/H_0$ in a Planck cosmology [12],

$$L \gtrsim 6\,\frac{c}{H_0} \tag{8}$$

as an order-of-magnitude statement (exact numerical factors depend on the statistic and priors). Recent topology surveys likewise place the shortest E_1 side near the diameter scale of the last-scattering surface, not one Hubble radius [6, 8].

Write $L = \alpha\,c/H_0$. Empirical scale matching uses $\alpha = 1$; cubic bounds require $\alpha \sim \mathcal{O}(6)$. If the phenomenological formula is reinterpreted as $a_0 = c^2/(2\pi L) = cH_0/(2\pi\alpha)$, topology-safe α yields

$$a_0 \lesssim \frac{1}{6} \times \frac{cH_0}{2\pi} \sim 1.8 \times 10^{-11}\,\mathrm{m\,s^{-2}}\,, \tag{9}$$

destroying the intended MOND-scale match. Therefore the cubic-E_1, $\alpha = 1$ package that simultaneously targets $a_0 \sim 10^{-10}$ and CMB non-detection is a no-go.

Vague statements that a "Hubble-scale domain is compatible with non-detection" are incorrect for cubic E_1.

V. PRESENT-EPOCH MATCHING VERSUS FIXED-COMOVING FLRW

For fixed comoving moduli $L^{(\rm com)}$,

$$L_{\rm phys}(t) = a(t)\,L^{(\rm com)}\,, \qquad \frac{\dot L_{\rm phys}}{L_{\rm phys}} = H\,. \tag{10}$$

Requiring instead $L_{\rm phys}(t) = c/H(t)$ gives

$$\frac{\dot L_{\rm phys}}{L_{\rm phys}} = -\frac{\dot H}{H}\,, \tag{11}$$

and equating the two yields $\dot H = -H^2$, i.e. deceleration parameter $q = 0$ (coasting). Exact tracking is therefore incompatible with ordinary matter-, radiation-, or Λ-dominated FLRW evolution under fixed comoving moduli.

The remaining alternatives are: (i) present-epoch numerical identification only; or (ii) an independent modulus sector with its own kinetic term and stress-energy. No such sector is constructed here.

The Hubble sphere c/H is also not generally a particle or event horizon; calling it a "global causal domain" is physically misleading [13].

VI. THE PROJECTOR RATIO DOES NOT DETERMINE $C_{\rm obs}$

A. The identity

For any unit n^i in three-dimensional Riemannian geometry,

$$h_{ij} = \gamma_{ij} - n_i n_j\,, \qquad \frac{\mathrm{Tr}(h)}{\mathrm{Tr}(\gamma)} = \frac{2}{3}\,. \tag{12}$$

This is correct, topology-independent, and independent of superfluidity. It discards tensorial orientation information and is not special to T^3.

B. Static AQUAL-type action

A static AQUAL-type force sector [7] may be written

$$S_{\rm WF} = \int dt\,d^3x\left[-\frac{|\nabla\Phi_N|^2}{8\pi G} - \rho_b\Phi_N - \frac{C_{\rm IR}}{12\pi G\,a_0}\,|\nabla\psi|^3 - C_m\,\rho_b\,\psi\right], \tag{13}$$

with $q = |\nabla\psi| = \sqrt{\gamma^{ij}\partial_i\psi\partial_j\psi}$. Neither n^i nor h_{ij} appears. The kinetic operator is fully three-dimensional and isotropic. There is therefore no field-theoretic sense in which Eq. (12) "matches" onto the EFT.

Variation yields

$$\nabla\cdot\left(\frac{|\nabla\psi|}{a_0}\,\nabla\psi\right) = 4\pi G\,\frac{C_m}{C_{\rm IR}}\,\rho_b\,. \tag{14}$$

In spherical symmetry, with $g_N = |\nabla\Phi_N|$ and physical acceleration $g_P = C_m|\nabla\psi|$,

$$g_P = C_{\rm obs}\sqrt{a_0\,g_N}\,, \qquad \boxed{C_{\rm obs} = \frac{C_m^{3/2}}{\sqrt{C_{\rm IR}}}}\,. \tag{15}$$

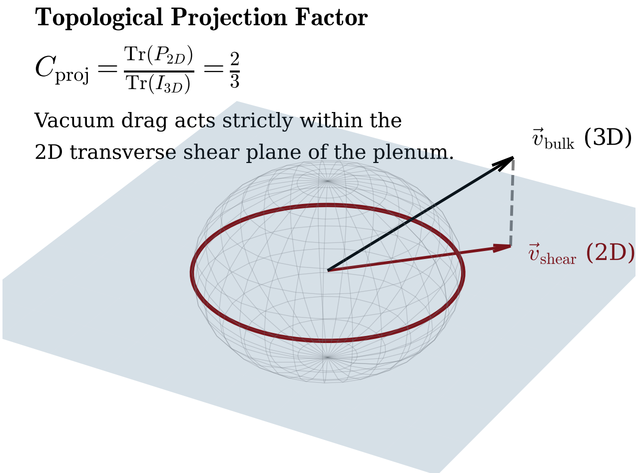

FIG. 2. Isotropic degree-of-freedom cartoon only. It does not appear in Eq. (13) and must not be read as a derivation of C_{obs}.

C. Field-rescaling invariance

Under $\psi \mapsto s\psi$ with $s > 0$,

$$C_m \mapsto \frac{C_m}{s}, \qquad C_{\text{IR}} \mapsto \frac{C_{\text{IR}}}{s^3}, \tag{16}$$

while C_{obs} is invariant. Thus C_{obs} is the unique combination of the two positive coefficients that can enter observables. Setting $C_m = C_{\text{IR}}$ so that $C_{\text{obs}} = C$ is a normalization convention, not a calculation of $C = 2/3$.

a. Withdrawn. Any statement that $C_{\text{obs}} = 2/3$ follows from T^3 geometry, from Eq. (12), or from "matching" without inserting h^{ij} into the action and integrating out fields.

D. Compact sources

On compact T^3, the integral of any pure divergence vanishes, so localized solutions must use a compensated density contrast (and, if present, a zero-mean exchange source). For general morphology,

$$\frac{|\nabla\varphi|}{a_0}\,\nabla\varphi = \frac{C_m}{C_{\text{IR}}}\,\nabla\Phi_N + \nabla\times\mathbf{H}, \tag{17}$$

so pointwise square-root formulae are approximations for disks. These statements remain valid; they do not resurrect $C_{\text{obs}} = 2/3$.

VII. MUTUAL INCONSISTENCY WITH THE EMPIRICAL RAR SCALE

In the deep-MOND regime, $g_P = C_{\text{obs}}\sqrt{a_0 g_N}$ is equivalent to

$$g_P = \sqrt{a_{0,\text{eff}}\, g_N}, \qquad a_{0,\text{eff}} = C_{\text{obs}}^2\, a_0\,. \tag{18}$$

Imposing both $a_0 = cH_0/2\pi \approx 1.1\times10^{-10}\,\text{m}\,\text{s}^{-2}$ and $C_{\text{obs}} = 2/3$ yields

$$a_{0,\text{eff}} = \frac{4}{9}a_0 \approx 4.9\times10^{-11}\,\text{m}\,\text{s}^{-2}, \tag{19}$$

whereas modern RAR determinations find $a_0 \simeq 1.2\times10^{-10}\,\text{m}\,\text{s}^{-2}$ [3, 4], a factor ~ 2.4 discrepancy. Equivalently, asymptotic baryonic Tully–Fisher velocities scale as $v^4 \propto C_{\text{obs}}^2 a_0 G M_b$; $C_{\text{obs}} = 2/3$ lowers v by $\sim 18\%$ at fixed baryonic mass, which is observationally large.

Therefore one cannot simultaneously advertise $a_0 = cH_0/2\pi$ as the empirical MOND scale and impose a universal $C_{\text{obs}} = 2/3$. Consistent options for any future fit are:

1. set $C_{\text{obs}} = 1$ (up to the definition of a_0);
2. redefine the input scale so that $C_{\text{obs}}^2 a_0$ matches the RAR;
3. drop universality and derive a morphology-dependent response from a solved disk PDE before fitting.

VIII. OUTLOOK: RETAINED RESULTS AND A RECONSTRUCTION PATH

A. What this note establishes

1. The coincidence (1) is empirical motivation only.
2. $a_0 \equiv cH_0/2\pi$ is allowed solely as a *present-epoch phenomenological postulate.*
3. The weak-field combination $C_{\text{obs}} = C_m^{3/2}/\sqrt{C_{\text{IR}}}$ is a derived, normalization-invariant observable coefficient [Eq. (15)].
4. On compact T^3, sources must be compensated and disk solutions generally admit a curl term [Eq. (17)].
5. The no-go results of Secs. III–VII rule out the usual cubic geometric packaging of those two numbers.

B. Path required before any geometric-derivation claim

A subsequent work that claims to *derive* a_0 or C_{obs} from topology or microphysics must close the following chain. Items are grouped by sector; each is a separate technical gate, not a slogan.

a. Microphysics and circulation.

(M1) Define a condensate order parameter $\Psi = \sqrt{\rho}\, e^{i\theta}$ (or a relativistic analogue) with a definite mass parameter m.

(M2) Obtain the circulation from $\nabla\theta$ and impose integer winding on non-contractible cycles of T^3, yielding $\Gamma = n\, h/m$ rather than $\Gamma = cL$.

(M3) Fix a single geometric meaning of L (cube edge, closed geodesic length, or circumference) and use it consistently in every formula.

(M4) Derive a map $(\Gamma, m, n, L) \mapsto a_0$ that does not rely on a chart-dependent "distribution" of cH_0 over an angular range of 2π.

b. Cosmic scale and moduli.

(S1) Either restrict scale matching to the present epoch, or supply a modulus action for $L_{\rm phys}(t)$ that does not force the coasting condition $q = 0$.

(S2) Confront the resulting L with Planck/COMPACT bounds for cubic E_1 and for twisted flat alternatives [6, 8].

c. Weak-field coefficient and data.

(E1) If a projector h_{ij} is essential, insert it into the action, specify the dynamics of n^i, and compute $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$ as a definite number independent of field normalization.

(E2) Reconcile $C_{\rm obs}^2 a_0$ with the empirical RAR/BTFR normalization (avoid the factor-4/9 conflict of Sec. VII).

(E3) Solve the compensated periodic nonlinear equation for at least one spherical and one disk baryonic source, including any curl contribution.

d. Representation (already satisfied here).

(R1) Use a face-identified parallelepiped for T^3 (Fig. 1), never an embedded doughnut T^2.

Until (M1)–(M4), (S1)–(S2), and (E1)–(E3) are closed, the defensible scientific product remains exactly what this note provides: present-epoch phenomenological matching of Eq. (1), the derived invariant $C_{\rm obs}$, and the no-go results for the cubic geometric story—not a geometric derivation of the MOND scale.

CODE AND DATA AVAILABILITY

Figure 1 is produced by `Scripts/itsm_t3_fundamental_domain.py` in the public archive `https://github.com/brendohxd/ITSM-Integrated-Toroidal-Syntropic-Model` (Zenodo DOI 10.5281/zenodo.18808348). Scripts that render an embedded doughnut surface depict T^2 and should not be used as T^3 schematics. No new observational catalogue is analysed.

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